

Bob's Pizza Emporium

Software Requirements Specification

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Version 2.0

The Quintessentials

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Revision History

Date	Version	Description	Author(s)
10/2/2025	1.0	Initial version	Joshua Swanson, Noah Allen, Alison Michailof, Kelly Byrne, Caelyn Ruiz
10/28/25	2.0	Second version	Joshua Swanson, Noah Allen, Alison Michailof, Kelly Byrne, Caelyn Ruiz

1. Introduction

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements for the Bob's Pizza Emporium application. This document shall serve as the basis for the system interface, software, and hardware requirements, and application performance. This will be used to create a Test plan that ensures all requirements are met.

1.2 Scope

This Software Requirements Specification will outline system requirements that will be used for this project. All work outlined in this document is written in full detail, based on the requirements interpreted at the time of writing.

1.3 Definitions, Acronyms, and Abbreviations

Acronym	Meaning
GUI	Graphical User Interface
PIN	Personal Identification Number
RAM	Random Access Memory
SDP	Software Development Plan
SRS	Software Requirements Specification

1.4 Document References

Document Title	Version	Date	Author(s)
Software Development Plan	1.0	9/18/2025	Joshua Swanson, Noah Allen, Alison Michailof, Kelly Byrne, Caelyn Ruiz

1.5 System Overview

The application shall be used by employees of Bob's Pizza Emporium to record customer orders, configure pizzas, and process order transactions. The system shall display the menu, add items, calculate the total bills with tax, and finalize transactions. Administrative users shall have access to advanced settings and configurations for the user accounts. The application shall be deployed as a Windows executable that is compatible with Windows operating systems.

2. Functional Description

The software requirements reflect the customer's description of the application. All Mandatory requirement statements within the document will include the word "shall." All optional requirement statements within the document will include the word "should." Non-requirement statements are any that were not specified in the initial description, and are only desired features.

2.1 Login Display

The user shall be presented with a login screen where the user shall enter a valid username and a 4-digit PIN.

2.1.1 Login Validation

The system shall validate the user's credentials. If the validation is a success, the system shall authenticate the user. If the validation is unsuccessful, the system shall deny access to the user and prompt them to re-enter their credentials.

2.1.2 Login Attempts

The user shall be able to attempt to log in an unlimited number of times.

2.1.3 Password Reset

The system should display a "Forgot Password" button that, when selected, shall alert an admin. The admin shall be able to reset the user's password.

2.1.4 Administrative Login

The admin shall be able to log in using an administrative 4-digit PIN.

The admin shall be able to create, destroy, and modify user accounts, including adding and removing new admins.

2.2 Main Display

The main screen shall display the name of the logged-in user at the top right of the screen.

The main screen shall display user information, menu items, order summary, and administrative options (if logged in as an admin).

2.2.1 User Display

The user shall be able to view standard pizza orders

The user shall be able to select a custom pizza option for personalized orders.

The system shall provide an “add to cart” option to allow items to be placed into an order.

The user shall be able to edit their order in the cart at any time before submitting the order.

The system shall provide pricing details for all items, including tax.

The system shall allow users to customize pizzas, select drinks, and edit orders before finalizing an order.

2.2.2 Admin Display

The admin shall be able to see only administrative menus and settings.

The admin shall be able to view all active user accounts.

2.3 Ordering and Menu

The menu shall include a standard set of pizzas with configurable options.

The system shall provide options for different-sized pizzas: small, medium, and large.

The system shall allow users to add toppings to their pizzas, including: Pepperoni, Sausage, Bacon, Pineapple, Mushrooms, and Onions.

The system shall provide a drinks menu.

The system shall only allow red sauce.

The system shall display pricing for all menu items.

The system shall configure price rates.

The system shall configure tax rates.

The system shall have persistent price and tax rate data when the program is closed.

The system shall calculate the order total.

The system shall have order completion options.

3. System Requirements

The section describes the minimum hardware and software requirements to run the pizza ordering application.

3.1 Hardware Requirements

The application shall be compatible with any computer that meets the following specifications:

The system shall require at least 4 GB of RAM.

The system shall require an active Internet or local network connection.

3.2 Software Requirements

The application shall be compatible with the following:

The operating system shall be Windows 10 or Windows 11.

The system shall support standard graphical libraries available on Windows.

4. Interfaces

The application includes human and system interface components.

4.1 Human Interface

The interface should display graphical menu icons.

4.2 System Interface

The application shall have persistent user account data when the program is closed.

The system should allow the admin to update menu options.

The application should store all orders in the database for reporting and future reference.

5. Performance

5.1 App Performance

5.1.1 GUI Navigation

The application should respond to user inputs within 1.0 seconds.

The application should focus on more optimized image formats rather than large graphical formats. This allows the use of image representation for a visually intuitive layout without using resource-heavy graphic formats that hinder performance (may not always load resource-heavy images).

5.1.2 Ordering Processing

The application should process and confirm orders within 2.0 seconds.

5.1.3 Administrative Actions

Administrative actions should be executed within 2.0 seconds.

6. Delivery

The following requirements address the delivery of Bob's Pizza Emporium.

6.1 Software Installation

The application shall be delivered as an installable package compatible with Windows 10 and Windows 11.

The installation package shall include all required dependencies to run the application.

6.2 Software Availability

The application shall possess all necessary components to function as an ordering system, including the menu display, login, order processing, and administrative features once installed.

6.3 Documentation

The Documentation shall be version-controlled and updated in parallel with software releases. Any bug fixes, feature additions, or interface design changes will be accompanied by document revisions.

7. Schedule

Artifacts shall be delivered on the following dates:

Artifact	Delivery Date
Software Requirements Specification	October 2, 2025
GUI Icon Prototype	October 9, 2025
GUI Menu Prototype	October 14, 2025
Revisions of GUI Prototypes	TBD

Final Software Version Ready for Deployment	TBD
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The updated SRS document version shall be delivered on October 28, 2025, as referenced in the SDP.

8. Miscellaneous

There are no miscellaneous items at this time.